# Chapter 4
## Sets

*Discrete Structures for Computing* on June 16, 2024

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1. Sets

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# Course outcomes

| | Course learning outcomes |
|---|---|
| L.O.1 | Understanding of logic and discrete structures |
| | L.O.1.1 – Describe definition of propositional and predicate logic |
| | L.O.1.2 – Define basic discrete structures: set, mapping, graphs |
| L.O.2 | Represent and model practical problems with discrete structures |
| | L.O.2.1 – Logically describe some problems arising in Computing |
| | L.O.2.2 – Use proving methods: direct, contrapositive, induction |
| | L.O.2.3 – Explain problem modeling using discrete structures |
| L.O.3 | Understanding of basic probability and random variables |
| | L.O.3.1 – Define basic probability theory |
| | L.O.3.2 – Explain discrete random variables |
| L.O.4 | Compute quantities of discrete structures and probabilities |
| | L.O.4.1 – Operate (compute/ optimize) on discrete structures |
| | L.O.4.2 – Compute probabilities of various events, conditional ones, Bayes theorem |



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# Set Definition

- Set is a **fundamental** discrete structure on which all discrete structures are built
- Sets are used to group objects, which often have the **same properties**

**Example**

- Set of all the students who are currently taking Discrete Mathematics 1 course.
- Set of all the subjects that K2011 students have to take in the first semester.
- Set of natural numbers $\mathbb{N}$

**Definition**

A set is an unordered collection of objects.
The objects in a set are called the elements (*phần tử*) of the set.
A set is said to contain (*chứa*) its elements.



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# Notations

**Definition**

- $a \in A$: $a$ is an element of the set $A$
- $a \notin A$: $a$ is not an element of the set $A$

**Definition (Set Description)**

- The set $V$ of all vowels in English alphabet, $V = \{a, e, i, o, u\}$
- Set of all real numbers greater than 1???
  $\{x \mid x \in \mathbb{R}, x > 1\}$
  $\{x \mid x > 1\}$
  $\{x \ : \ x > 1\}$



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# Equal Sets

**Definition**

Two sets are equal iff they have the same elements.

- $(A = B) \leftrightarrow \forall x(x \in A \leftrightarrow x \in B)$

**Example**

- $\{1, 3, 5\} = \{3, 5, 1\}$
- $\{1, 3, 5\} = \{1, 3, 3, 3, 5, 5, 5, 5\}$



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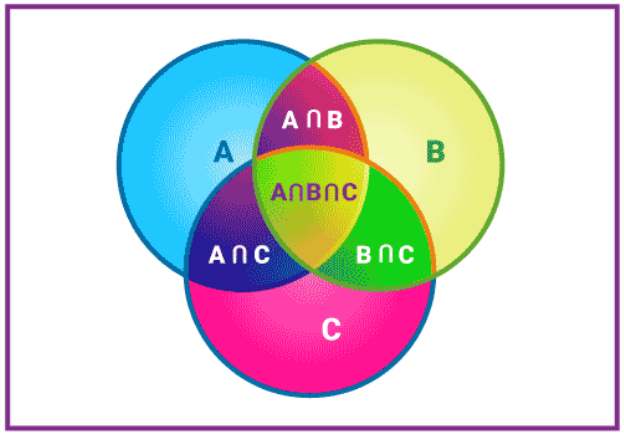
# Venn Diagram

- John Venn in 1881
- Universal set (*tập vũ trụ*) is represented by a rectangle
- Circles and other geometrical figures are used to represent sets
- Points are used to represent particular elements in set



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# Special Sets

- Empty set (*tập rỗng*) has no elements, denoted by $\emptyset$, or $\{\}$
- A set with one element is called a singleton set
- What is $\{\emptyset\}$?
- Answer: singleton



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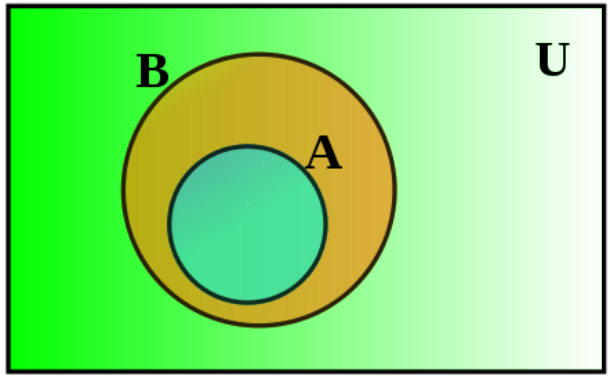
# Subset

**Definition**

The set $A$ is called a subset (*tập con*) of $B$ iff every element of $A$ is also an element of $B$, denoted by $A \subseteq B$.

If $A \neq B$, we write $A \subset B$ and say $A$ is a proper subset (*tập con thực sự*) of $B$.

- $\forall x(x \in A \rightarrow x \in B)$
- For every set $S$,
  (i) $\emptyset \subseteq S$, (ii) $S \subseteq S$.



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# Cardinality

**Definition**

If $S$ has exactly $n$ distinct elements where $n$ is non-negative integers, $S$ is finite set (*tập hữu hạn*), and $n$ is **cardinality** (*bản số*) of $S$, denoted by $|S|$.

**Example**

- $A$ is the set of odd positive integers less than 10. $|A| = 5$.
- $S$ is the letters in Vietnamese alphabet, $|S| = 29$.
- Null set $|\emptyset| = 0$.

**Definition**

A set that is infinite if it is not finite.

**Example**

- Set of positive integers is infinite



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# Power Set

**Definition**

Given a set $S$, the power set (*tập lũy thừa*) of $S$ is the set of all subsets of the set $S$, denoted by $P(S)$.

**Example**

What is the power set of $\{0, 1, 2\}$?
$P(\{0, 1, 2\}) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$

**Example**

- What is the power set of the empty set?
- What is the power set of the set $\{\emptyset\}$

# Power Set

**Theorem**

*If a set has $n$ elements, then its power set has $2^n$ elements.*

Prove using induction!



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# Ordered $n$-tuples

**Definition**

The ordered $n$-tuple (*dãy sắp thứ tự*) $(a_1, a_2, \ldots, a_n)$ is the ordered collection that has $a_1$ as its first element, $a_2$ as its second element, ..., and $a_n$ as its $n$th element.

**Definition**

Two ordered $n$-tuples $(a_1, a_2, \ldots, a_n) = (b_1, b_2, \ldots, b_n)$ iff $a_i = b_i$, for $i = 1, 2, \ldots, n$.

**Example**

2-tuples, or **ordered pairs** (*cặp*), $(a, b)$ and $(c, d)$ are equal iff $a = c$ and $b = d$



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# Cartesian Product

- René Descartes (1596–1650)

**Definition**

Let $A$ and $B$ be sets. The Cartesian product (*tích Đề-các*) of $A$ and $B$, denoted by $A \times B$, is the set of ordered pairs $(a, b)$, where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

**Example**

Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$. Then

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

Show that $A \times B \neq B \times A$



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# Cartesian Product

## Definition

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \ldots, n\}$$

## Example

$A = \{0, 1\}, B = \{1, 2\}, C = \{0, 1, 2\}$. What is $A \times B \times C$?

$$\begin{aligned} A \times B \times C &= \{(0,1,0), (0,1,1), (0,1,2), (0,2,0), (0,2,1), \\ &\quad (0,2,2), (1,1,0), (1,1,1), (1,1,2), (1,2,0), \\ &\quad (1,2,1), (1,2,2)\} \end{aligned}$$



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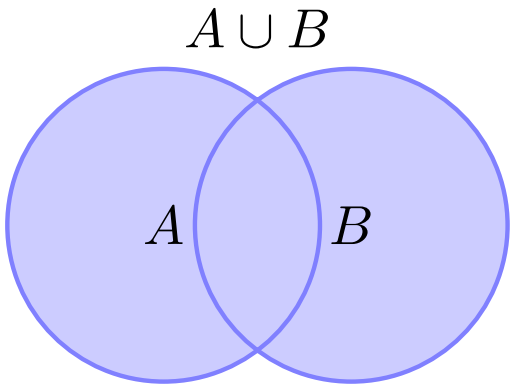
# Union

**Definition**

The union (*hợp*) of A and B

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$



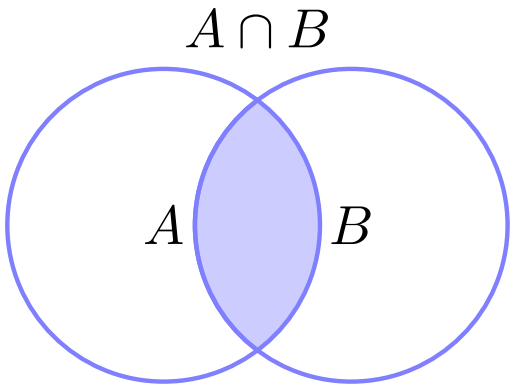

- Example:
  - $\{1,2,3\} \cup \{2,4\} = \{1,2,3,4\}$
  - $\{1,2,3\} \cup \emptyset = \{1,2,3\}$

# Intersection

**Definition**

The intersection (*giao*) of A and B

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$




Example:

- $\{1,2,3\} \cap \{2,4\} = \{2\}$
- $\{1,2,3\} \cap \mathbb{N} = \{1,2,3\}$



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# Union/Intersection

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \ldots \cup A_n = \{x \mid x \in A_1 \vee x \in A_2 \vee \ldots \vee x \in A_n\}$$

$$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \ldots \cap A_n = \{x \mid x \in A_1 \wedge x \in A_2 \wedge \ldots \wedge x \in A_n\}$$



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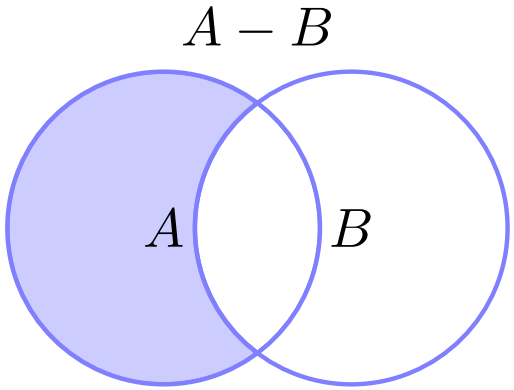
# Difference

**Definition**

The difference (*hiệu*) of A and B

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$




Example:

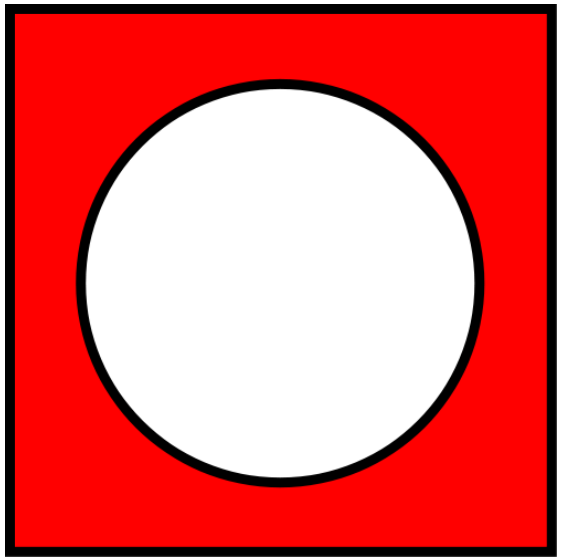
- {1,2,3} - {2,4} = {1,3}
- {1,2,3} - $\mathbb{N}$ = $\emptyset$

# Complement

**Definition**

The complement (*phần bù*) of A

$$\overline{A} = \{x \mid x \notin A\}$$



Example:

- A = {1,2,3} then $\overline{A}$ = ???
- **Note that A - B = A $\cap \overline{B}$**

# Set Identities

| | | | |
|---|---|---|---|
| $A \cup \emptyset$ | $=$ | $A$ | Identity laws |
| $A \cap U$ | $=$ | $A$ | Luật đồng nhất |
| $A \cup U$ | $=$ | $U$ | Domination laws |
| $A \cap \emptyset$ | $=$ | $\emptyset$ | Luật nuốt |
| $A \cup A$ | $=$ | $A$ | Idempotent laws |
| $A \cap A$ | $=$ | $A$ | Luật lũy đẳng |
| $\overline{(\bar{A})}$ | $=$ | $A$ | Complementation law |
| | | | Luật bù |

# Set Identities

| | | | |
|---|---|---|---|
| $A \cup B$ | $=$ | $B \cup A$ | Commutative laws |
| $A \cap B$ | $=$ | $B \cap A$ | Luật giao hoán |
| $A \cup (B \cup C)$ | $=$ | $(A \cup B) \cup C$ | Associative laws |
| $A \cap (B \cap C)$ | $=$ | $(A \cap B) \cap C$ | Luật kết hợp |
| $A \cup (B \cap C)$ | $=$ | $(A \cup B) \cap (A \cup C)$ | Distributive laws |
| $A \cap (B \cup C)$ | $=$ | $(A \cap B) \cup (A \cap C)$ | Luật phân phối |
| $\overline{A \cup B}$ | $=$ | $\overline{A} \cap \overline{B}$ | De Morgan's laws |
| $\overline{A \cap B}$ | $=$ | $\overline{A} \cup \overline{B}$ | Luật De Morgan |



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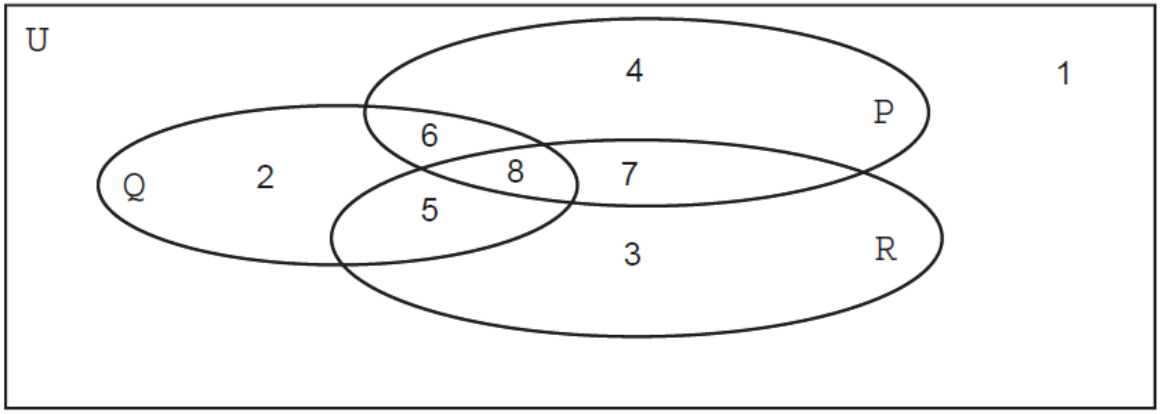
# Method of Proofs of Set Equations

To prove $A = B$, we could use

- Venn diagrams
- Prove that $A \subseteq B$ and $B \subseteq A$
- Use membership table
- Use set builder notation and logical equivalences

# Example (1)




## Example

Verify the distributive rule $P \cup (Q \cap R) = (P \cup Q) \cap (P \cup R)$

# Example (2)

## Example

Prove: $\overline{A \cap B} = \overline{A} \cup \overline{B}$

(1) Show that $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$

Suppose that $x \in \overline{A \cap B}$
By the definition of complement, $x \notin A \cap B$
So, $x \notin A$ or $x \notin B$
Hence, $x \in \bar{A}$ or $x \in \bar{B}$
We conclude, $x \in \overline{A} \cup \overline{B}$
Or, $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$

(2) Show that $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

# Example (3)

Prove: $\overline{A \cap B} = \overline{A} \cup \overline{B}$

| $A$ | $B$ | $A \cap B$ | $\overline{A \cap B}$ | $\bar{A} \cup \bar{B}$ |
|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 |

# Example (4)

Prove: $\overline{A \cap B} = \overline{A} \cup \overline{B}$

$$
\begin{aligned}
\overline{A \cap B} &= \{x | x \notin A \cap B\} \\
&= \{x | \neg(x \in A \cap B)\} \\
&= \{x | \neg(x \in A \wedge x \in B)\} \\
&= \{x | \neg(x \in A) \vee \neg(x \in B)\} \\
&= \{x | x \notin A \vee x \notin B\} \\
&= \{x | x \in \overline{A} \vee x \in \overline{B}\} \\
&= \{x | x \in \overline{A} \cup \overline{B}\}
\end{aligned}
$$